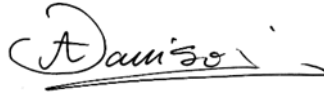


Approved By Associate Dean:



Jan 2, 2025

Signature

COURSE SECTION INFORMATION

Machine Learning II Applied A.I. Solutions Development

Note: All academic inquiries will be replied to within three business days.

COURSE DESCRIPTION:

This course will provide an in-depth overview of Machine Learning using NLP and Computer Vision. The main focus of this course will be on building NLP models & Computer Vision models for text, image and video analysis. Students will gain experience with professional tools and techniques such as Scikit-learn, NLTK, Spacy, Librosa, and PyAudioAnalysis and OpenCV.

COURSE OUTCOMES:

Upon successful completion of this course the students will have reliably demonstrated the ability to:

1. Assess, engineer, and select the appropriate data attributes (features) for building Machine Learning models for text, image and video use-cases
2. Apply image analysis techniques such as Object detection, Object Segmentation, Object Classification
3. Apply video analysis techniques such as Object Tracking, Object Localization
4. Build Machine Learning models using Scikit-learn
5. Build NLP and Computer Vision models using OpenCV and NLTK
6. Evaluate and finetune appropriate NLP and Computer Vision models

LIST OF TEXTBOOKS AND OTHER TEACHING AIDS:

Required:

Recommended Resources:

1. **Practical Natural Language Processing - Sowmya & et al.**
2. **Digital Image Processing 4th edition - Rafael Gonzalez & Richard Woods**

COURSE DELIVERY MODE:

Refer to the table below for the delivery mode.

Important Note on the Use of Generative AI:

Students must review the "Generative AI Usage Guidelines" document, available on D2L, or consult with the instructor for details on how generative AI tools may be used in this course.

Generally, use of AI is allowed and encouraged in most of the courses of this program, this include projects and assignments. However use of AI are not allowed in exams and quizzes unless specified otherwise by the instructor.

Students must consult their instructor when unsure.

Misuse of AI in assessments where it is not permitted or failure to adequately disclose its use will be treated as a violation of academic integrity. According to college policy, consequences may include failing the assignment or the course or more severe disciplinary actions. **Students must also download the AI Usage Declaration form from D2L, complete it, and submit it with their assignments where AI use is permitted.** Adherence to these guidelines is mandatory to maintain academic integrity.

Detailed Evaluation System

Assessment Tool:	Description:	Outcome(s) assessed:	EES assessed:	Date / Week:	% of Final Grade:
Participation	Attendance and in-class participation in various activities	1,2,3,4,5,6,7	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11	1 - 4	10
Lab Exercises 15 @ 2% (best 15 out of 20)	Hands-on exercises	1,2,3,4,5,6,7	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11	1 - 4	30
Project	Individual Assignments	1,2,3,4,5,6,7	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11	4	30
Final Exam	This is a short answer exam that will evaluate the knowledge of the student	1,2,3,4,5,6,7	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11	4	30
TOTAL:					100%

GRADING SYSTEM the passing grade for this course is: D (50%)

A+	90-100	4.0	B+	77-79	3.3	C+	67-69	2.3	D+	57-59	1.3	Below 50	F	0.0
A	86-89	4.0	B	73-76	3.0	C	63-66	2.0	D	50-56	1.0			
A-	80-85	3.7	B-	70-72	2.7	C-	60-62	1.7						

Excerpt from the College Policy on Academic Dishonesty:
The *minimal* consequence for submitting a plagiarized, purchased, contracted, or in any manner inappropriately negotiated or falsified assignment, test, essay, project, or any evaluated material will be a grade of zero on that material.

To view George Brown College policies please go to www.georgebrown.ca/policies

Learning Schedule / Topical Outline (subject to change with notification)

TOPICAL OUTLINE:

Week	Topic Task	In-person/Online	Outcome(s)	Content / Activities	Resources
1	1	Monday Online	1, 4, 5	● Introduction to NLP ● NLP Pipelines	Resource material available on D2L
1	2	Wednesday Online	1, 4, 5	• Feature Extraction in Text • Text Classification	Resource material available on D2L
1	3	Friday Online	1, 4, 5	• Knowledge Extraction • Topic Modeling	Resource material available on D2L
Tasks: Lab Exercises 1, 2, 3, 4					
2	4	Monday In-person	1, 2, 4, 5, 6	• Recommenders	Resource material available on D2L
2	5	Wednesday In-person	1, 2, 4, 5, 6	• Summarizers • Chatbots	Resource material available on D2L

2	6	Friday Online	1, 2, 4, 5, 6	• Introduction to Computer Vision using OpenCV • Fundamentals of Image processing	Resource material available on D2L
Tasks: Lab Exercises 5, 6, 7, 8, 9, 10					
3	7	Monday In-person	1, 2, 3, 4, 5, 6	• Transformations ● Histograms ● Convolutions	Resource material available on D2L
3	8	Wednesday In-person	1, 2, 3, 4, 5, 6	● Image Smoothing ● Thresholding ● Edge detection ● Contours	Resource material available on D2L
3	9	Friday Online	1, 2, 3, 4, 5, 6	● Fundamentals of Video processing • Motion detection • Object Masking	Resource material available on D2L
Tasks: Lab Exercises 11, 12, 13, 14, 15, 16					
4	10	Monday In-person	1, 2, 3, 4, 5, 6	• Introduction to Audio Processing ● Tour of PyAudioAnalysis, Librosa ● Feature Extraction in Audio	Resource material available on D2L
4	11	Wednesday Online	1, 2, 3, 4, 5, 6	● Audio Classification ● Speaker Identification ● Speech-To-Text Conversion ● Music Synthesis	Resource material available on D2L
4	12	Friday In-person	1, 2, 3, 4, 5, 6	● Final Exam	Resource material available on D2L
Tasks: Lab Exercises 17, 18, 19, 20 Project Due Final Exam					
Please note: this schedule may change as resources and circumstances require. For information on withdrawing from this course without academic penalty, please refer to the College Academic Calendar: http://www.georgebrown.ca/Admin/Registr/PSCal.aspx					